

22/07/2026

* Linear regression: (X_i, Y_i) , $i = 1, 2, \dots, n$ → Model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$

→ Assumptions:

Identifiability: $E(\varepsilon_i | X = X_i) = 0$ for $i = 1, \dots, n$ Usual: $\text{Var}(\varepsilon_i | X = X_i) = \sigma^2$ for $i = 1, \dots, n$ (more generally $\sigma^2(X_i)$)Uncorrelated: $\text{Cov}(\varepsilon_i, \varepsilon_j | \{X_k\}_{k=1}^n) = 0$ for all $i \neq j$.→ Least squares (LS) estimates: $(\hat{\beta}_0, \hat{\beta}_1)$

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}; \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

• Fitted value: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ → estimates of $E(Y | X = X_i)$ - Residual: $e_i = Y_i - \hat{Y}_i = \hat{\varepsilon}_i$ (Note: $\bar{e} = 0$, $\sum e_i X_i = 0$?)- Est. of σ^2 : $\hat{\sigma}^2 = \sum_{i=1}^n e_i^2$ $\underbrace{\quad}_{n-2}$ degrees of
freedomof $\underline{e} = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$ Under uncorrelatedness of the ε_i 's, $\hat{\sigma}^2$ is unbiased for σ^2 .

→ LS v/s Normal likelihood:

Gaussian linear regression model: $Y_i | X_i$ are indep. $N(\beta_0 + \beta_1 X_i, \sigma^2)$, log likelihood of $(\beta_0, \beta_1, \sigma^2)$

given data:

$$L(\beta_0, \beta_1, \sigma^2 | (\underline{X}, \underline{Y})) = C_n - \frac{1}{2\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)^2$$

$$MLE: (\hat{\beta}_0^{ML}, \hat{\beta}_1^{ML}) = (\hat{\beta}_0^{LS}, \hat{\beta}_1^{LS}) - \frac{n}{2} \log \sigma^2$$

$$\hat{\sigma}^2^{ML} = \frac{1}{n} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 = \frac{1}{n} \sum e_i^2 = \frac{n-2}{n} \hat{\sigma}^2$$

• Statistics: $\bar{X}, \bar{Y}, S_{xy}, S_x^2, S_y^2$

Verify: SSE (sum of squared errors)

$$= \sum e_i^2 = \sum (Y_i - \bar{Y})^2 - \hat{\beta}_1^2 \sum (X_i - \bar{X})^2$$

$$- \hat{\beta}_1^2 \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n \{ \hat{\beta}_1 (X_i - \bar{X}) \}^2$$

$$= \sum_{i=1}^n \{ (\hat{\beta}_0 + \hat{\beta}_1 X_i) - (\hat{\beta}_0 + \hat{\beta}_1 \bar{X}) \}^2$$

$$= \sum_{i=1}^n \{ \hat{Y}_i - \bar{\hat{Y}} \}^2 \rightarrow \text{variability of the fitted value}$$

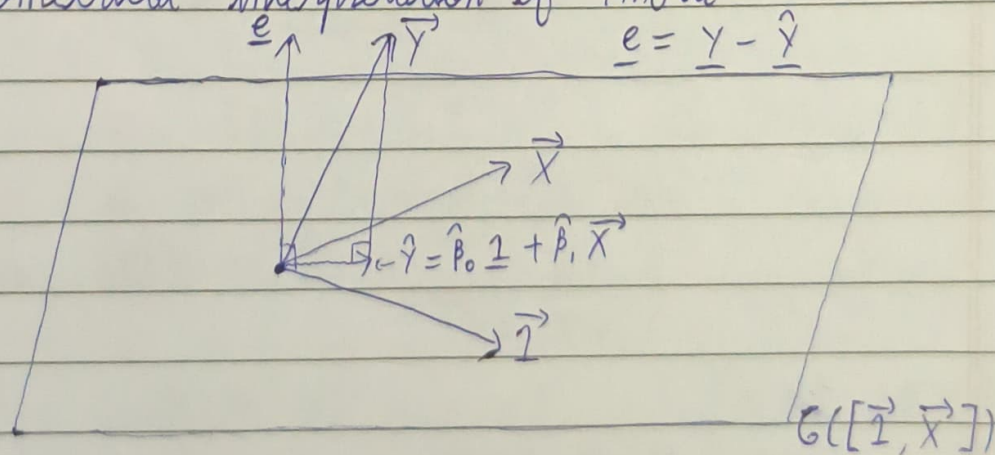
$$= \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 = SSR \text{ (sum of squares due to regression)}$$

$$\therefore SSE = \sum (Y_i - \bar{Y})^2 - SSR$$

$$\Rightarrow \sum (Y_i - \bar{Y})^2 = SSE + SSR$$

$$\Rightarrow SST = SSE + SSR \text{ (Anova decomposition)}$$

→ Geometrical interpretation of Anova



$$\underline{e}^T \underline{1} = 0 \text{ \& } \underline{e}^T \underline{X} = 0 \text{ (Normal equation)}$$

By Pythagoras theorem:

$$\|\underline{Y}\|^2 = \|\hat{\underline{Y}}\|^2 + \|\underline{e}\|^2$$

$$\Rightarrow \|\underline{Y}\|^2 - n\bar{Y}^2 = \|\hat{\underline{Y}}\|^2 - n\bar{Y}^2 + \|\underline{e}\|^2$$

$$\Rightarrow SST = (\|\hat{\underline{Y}}\|^2 - n\bar{Y}^2) + SSE$$

$$\Rightarrow SST = SSR + SSE$$

* Hat matrix:

$$\hat{\underline{Y}} = \hat{\beta}_0 \underline{1} + \hat{\beta}_1 \underline{X} = H \underline{Y}$$

$$H = \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T, \text{ here } \underline{X}_{n \times 2} = [\underline{1} : \underline{X}]$$

Some prop. of H:

• H is symmetric

• $HH = H$ (idempotence)

• $H\underline{1} = \underline{1}$ (b'coz $H\underline{X} = \underline{X}$)

$$\text{Let } P_1 = \frac{1}{n} \underline{1} \underline{1}^T \Rightarrow HP_1 = P_1$$

→ Quality of estimator: [Under assumⁿ ①-③]

Proposition a) $(\hat{\beta}_0, \hat{\beta}_1)$ is an unbiased estimate for (β_0, β_1)

b) $\hat{\sigma}^2$ " " " " " σ^2

Proof: a) $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (X^T X)^{-1} X^T Y$

$$E(\hat{\beta}) = (X^T X)^{-1} X^T E(Y)$$

$$= (X^T X)^{-1} X^T X \beta = \beta //$$

$$b) E(SSE) = E(\|(I-H)Y\|^2) \quad (Y = X\beta + \varepsilon, (I-H)X = 0)$$

$$= E(\|(I-H)\varepsilon\|^2)$$

$$= E(\varepsilon^T (I-H) \varepsilon)$$

$$= E[\text{trace}(\varepsilon^T (I-H) \varepsilon)]$$

$$= E[\text{trace}((I-H) \varepsilon \varepsilon^T)]$$

$$= \text{trace}((I-H) E(\varepsilon \varepsilon^T)) \rightarrow \sigma^2 I$$

$$= \sigma^2 \text{trace}(I-H)$$

$$= \sigma^2 p(I-H)$$

$$= \sigma^2 (n-2) //$$



Propⁿ: $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Proof $\text{Var}(\hat{\beta}) = \text{Var}(\underbrace{(X^T X)^{-1} X^T}_{C} Y)$

$$= C \text{Var}(Y) C^T \quad (\text{Var}(Y) = \sigma^2 I)$$

$$= \sigma^2 C C^T = \sigma^2 (X^T X)^{-1}$$

$$\text{Now, } X^T X = \begin{pmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{pmatrix}$$

$$\Rightarrow (X^T X)^{-1} = \frac{1}{\underbrace{n \sum X_i^2 - (\sum X_i)^2}_{n(\sum (X_i - \bar{X})^2)}} \begin{pmatrix} \sum X_i^2 & -\sum X_i \\ -\sum X_i & n \end{pmatrix}$$

$$\therefore \text{Var}(\hat{\beta}_0) = \sigma^2 \frac{\sum X_i^2}{n S_x^2} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_x^2} \right)$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_x^2}$$

Calculate $\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{\sum e_i^2}{n-2}\right)$

Propⁿ: If $Y_i | X_i \stackrel{\text{inde}}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2) \Rightarrow \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Then ~~(n-2)~~ $SSE = (n-2) \hat{\sigma}^2 \sim \sigma^2 \chi_{n-2}^2$

Proof:

$$SSE = \varepsilon^T (I - H) \varepsilon$$

$$= \varepsilon^T P \wedge P^T \varepsilon$$

$$= U^T \wedge U$$

$\therefore \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, $U = P^T \varepsilon \sim N(0, \sigma^2 I)$

$$\therefore U^T \wedge U = \sigma^2 \left(\sum_{i=1}^n \frac{U_i^2}{\sigma^2} \right) \sim \chi_{n-2}^2$$